

Model Answer Solution Paper

Subject Title: Artificial Intelligence & Machine Learning Algorithms

Subject Code: 315330 | **Time:** 3 Hours | **Max Marks:** 70

1. Attempt any FIVE of the following (10 Marks)

a) List any four applications of AI.

- Healthcare & Medical Diagnosis
- Autonomous Vehicles
- Natural Language Processing (NLP)
- Finance & Fraud Detection

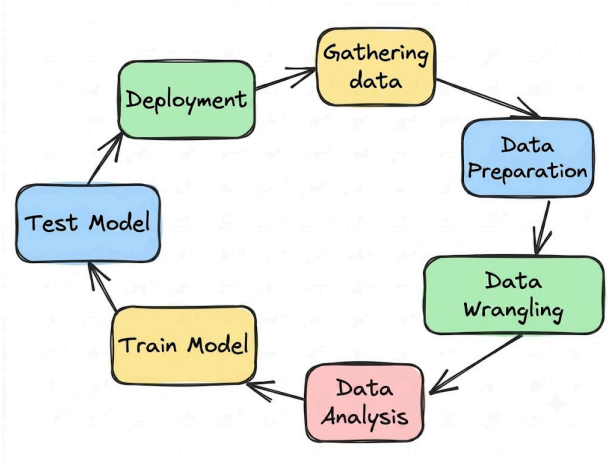
b) State the need for a search algorithm in Artificial Intelligence.

Search algorithms are fundamental in Artificial Intelligence because problem-solving in complex domains requires exploring a state space of possible decisions to find a sequence of actions that leads from an initial state to a desired goal state. They enable AI agents to evaluate alternative choices, optimize path costs, and solve problems where direct analytical or deterministic mathematical solutions do not exist (such as pathfinding, game playing, resource scheduling, and automated planning).

c) Compare forward chaining and backward chaining in AI.

Feature	Forward Chaining	Backward Chaining
Reasoning Direction	Data-driven: starts from known facts and applies inference rules to derive new facts.	Goal-driven: starts from a target hypothesis/goal and works backward to find supporting facts.
Strategy	Bottom-up processing approach.	Top-down processing approach.
Ideal Context	Efficient when data/facts are abundant and potential outcomes/goals are open-ended.	Efficient when the target goal is specific (e.g., medical diagnosis or system troubleshooting).

d) Draw machine learning life cycle.



e) Define unsupervised learning with example.

Unsupervised learning is a machine learning paradigm where an algorithm is provided with unlabeled input data without explicit target outputs. **Example:** Customer Segmentation using K-Means Clustering, where an e-commerce platform groups customers into distinct clusters based on browsing and purchase history without predefined demographic labels.

f) List metrics for classification.

- Classification Accuracy
- Precision
- Recall (Sensitivity or True Positive Rate)
- F1-Score (Harmonic Mean of Precision and Recall)
- ROC-AUC (Receiver Operating Characteristic - Area Under Curve)
- Confusion Matrix

g) Define linear regression with example.

Definition: Linear regression is a supervised learning algorithm used to model the linear relationship between a continuous scalar dependent variable (y) and one or more independent predictor variables (x) using a linear equation:

$$y = \beta_0 + \beta_1 x + \varepsilon$$

Example: Predicting house prices (y) based on the size of the house in square feet (x).

2. Attempt any THREE of the following (12 Marks)

a) Describe properties of A* algorithm.

A* is an informed search algorithm that evaluates nodes using the evaluation function $f(n) = g(n) + h(n)$, where $g(n)$ is the exact cost to reach node n from the start state, and $h(n)$ is an estimated heuristic cost from n to the goal.

- **Completeness:** A* is complete; it will guaranteed find a solution if one exists, provided the branching factor is finite and path step costs are strictly positive ($\varepsilon > 0$).
- **Admissibility (Optimality):** A* guarantees an optimal solution if the heuristic $h(n)$ is admissible—meaning $h(n)$ never overestimates the actual minimal cost to reach the goal ($h(n) \leq h^*(n)$).
- **Consistency (Monotonicity):** If $h(n)$ satisfies $h(n) \leq c(n, a, n') + h(n')$ for every node n and successor n' , A* expands nodes in monotonically increasing order of $f(n)$, ensuring no node requires re-evaluation.
- **Complexity:** Time and space complexity are exponential in the worst case, $O(b^d)$, where b is the branching factor and d is solution depth, because A* keeps all generated nodes in memory ('OPEN' and 'CLOSED' lists).

b) Illustrate Bayes' Theorem with example.

Bayes' Theorem computes the conditional probability of an event A given prior knowledge of conditions related to event B :

$$P(A | B) = [P(B | A) \times P(A)] / P(B)$$

- $P(A | B)$: Posterior probability of hypothesis A given evidence B .
- $P(B | A)$: Likelihood of observing evidence B given hypothesis A .
- $P(A)$: Prior probability of hypothesis A .
- $P(B)$: Marginal probability of evidence B .

Numerical Example:

Suppose a medical condition affects 1% of the population: $P(D) = 0.01$, $P(\neg D) = 0.99$. A diagnostic test has a 95% true positive rate ($P(T^+ | D) = 0.95$) and a 5% false positive rate ($P(T^+ | \neg D) = 0.05$).

1. Calculate total probability of a positive test $P(T^+)$:

$$P(T^+) = P(T^+ | D)P(D) + P(T^+ | \neg D)P(\neg D)$$

$$P(T^+) = (0.95 \times 0.01) + (0.05 \times 0.99) = 0.0095 + 0.0495 = 0.0590$$
2. Calculate $P(D | T^+)$ (probability patient has the disease given a positive test result):

$$P(D | T^+) = [P(T^+ | D) \times P(D)] / P(T^+) = 0.0095 / 0.0590 \approx 0.1610 \text{ (16.1\%)}$$

c) Differentiate between Training dataset and Testing dataset.

Feature	Training Dataset	Testing Dataset
Purpose	Used to fit and optimize parameters (weights and biases) of the machine learning model during learning.	Used to evaluate unbiased model performance and accuracy on unseen real-world data.
Data Split Ratio	Consumes the majority of the original dataset (typically 70% to 80%).	Consumes the remaining split portion (typically 20% to 30%).
Visibility to Model	Directly exposed to and processed by the learning algorithm during training iterations.	Kept completely hidden during training to prevent data leakage and bias.
Performance Metric	Measures model convergence, parameter fitting, and training error.	Measures generalization ability, variance, and true test error.

d) Explain positive and negative class cross-validation.

In binary classification tasks, samples are divided into two target groups: the **Positive Class** (the target or rare event of interest, e.g., disease detected) and the **Negative Class** (the normal background majority event, e.g., healthy patient).

Original Dataset Distribution: (20% Positive, 80% Negative)

Stratified 3-Fold Cross-Validation Folds:

Fold 1: [++ | - - - -] (20% Positive, 80% Negative)

Fold 2: [++ | - - - -] (20% Positive, 80% Negative)

Fold 3: [+ | - - - - -] (20% Positive, 80% Negative)

- **The Challenge:** Standard unstratified random cross-validation can partition dataset folds with skewed distributions of positive and negative classes, leading to biased training folds or testing folds lacking positive instances entirely.
- **Stratified Cross-Validation:** Solves class distribution imbalance by ensuring that every cross-validation fold preserves the exact proportion of positive and negative class samples as present in the full dataset.
- **Execution Steps:**
 1. Divide the complete dataset into K equal subsets while maintaining target class proportions.
 2. Train the model on $K - 1$ stratified folds and validate on the remaining fold.
 3. Repeat K times and compute average classification metrics (Precision, Recall, F1-Score) across all runs.

3. Attempt any THREE of the following (12 Marks)

a) Explain local search algorithm with example.

Local search algorithms operate on a single current state (rather than maintaining multiple systematic search paths) and move iteratively to neighboring states in the search space to optimize an objective function. They require $O(1)$ constant memory and are ideal for pure optimization problems where the final state matters rather than the path taken.

Example (Hill Climbing Algorithm):

1. Start with an initial candidate solution (e.g., random placement of 8 queens on an 8x8 chessboard).
2. Evaluate the current state using an objective function (e.g., number of conflicting pairs of queens).
3. Generate neighboring states by moving one queen to a new row within its column.
4. Move to the neighbor state that yields the maximum improvement in the objective function score.
5. Terminate when no adjacent neighbor state offers a better evaluation score (reaching a local peak/optimum).

b) Differentiate between Data Analytics and Data Science.

Feature	Data Analytics	Data Science
Scope	Focused on examining historical datasets to answer specific, targeted business queries.	Multidisciplinary field covering data pipeline design, predictive modeling, machine learning, and AI algorithms.
Core Objective	Diagnostic and descriptive analysis (understanding past trends and performance).	Predictive and prescriptive analysis (building models to forecast future events and automate decisions).
Primary Tools	SQL, Excel, Tableau, Power BI, R, Python (Pandas).	Python, R, TensorFlow, PyTorch, Apache Spark, Scikit-Learn.
Questions Addressed	"What happened in our business?" and "Why did it happen?"	"What will happen next?" and "How can we automatically optimize outcomes?"

c) Describe logistic regression learning algorithm.

Logistic regression is a supervised classification algorithm used to estimate the probability that a given input instance belongs to a binary categorical class.

- Sigmoid Activation Function:** Passes a linear combination of inputs through the logistic sigmoid function to map predicted values between 0 and 1:

$$\sigma(z) = 1 / [1 + e^{-z}] \quad \text{where} \quad z = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$
- Decision Boundary:**
 Output class is assigned as 1 if $\sigma(z) \geq 0.5$; otherwise assigned as 0 if $\sigma(z) < 0.5$.
- Loss Function (Binary Cross-Entropy / Log Loss):**

$$J(\beta) = - (1 / m) \sum [y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$$
 Model parameters (β) are updated iteratively using Gradient Descent to minimize loss $J(\beta)$.

d) Explain Binary and Multiclass classification.

- Binary Classification:**
 - Categorizes input instances into **two discrete, mutually exclusive classes** (0 or 1).
 - Examples: Email Spam Detection (Spam vs. Not Spam), Medical Diagnosis (Positive vs. Negative).
 - Common Algorithms: Logistic Regression, Binary Support Vector Machines (SVM), Decision Trees.
- Multiclass Classification:**
 - Categorizes input instances into **more than two distinct classes**.
 - Examples: Optical Character Recognition (Digits 0 through 9), Document Categorization (Sports, Politics, Tech, Finance).
 - Common Approaches: Native Multiclass (Softmax Regression, Random Forest,

Naive Bayes) or Binarization Strategies (One-vs-Rest, One-vs-One).

4. Attempt any THREE of the following (12 Marks)

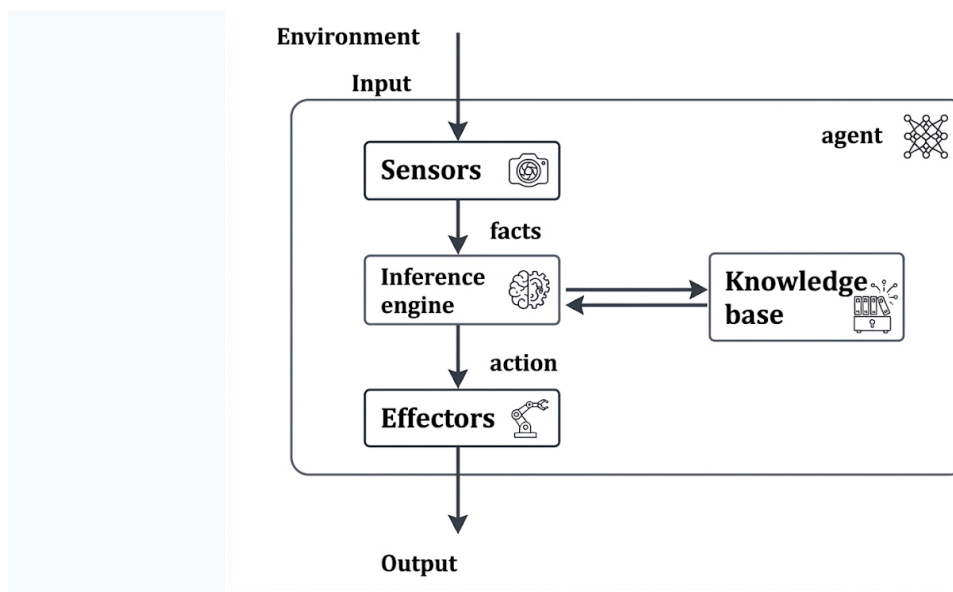
a) Explain Greedy Best-First technique with example.

Greedy Best-First Search is an informed search algorithm that expands the node estimated to be closest to the goal state based solely on its heuristic value: $f(n) = h(n)$.

- **Working Principle:** It maintains a priority queue of generated nodes ordered by $h(n)$. It expands the node with the lowest heuristic score without considering accumulated path cost $g(n)$.
- **Properties:** It is fast and memory efficient, but **neither optimal nor complete** (it can become trapped in infinite loops if state repetitions are not handled).
- **Example:**
Navigating a route from Start City S to Goal City G where heuristic $h(n)$ represents straight-line distance to G . From S , adjacent cities A ($h = 100$ km) and B ($h = 60$ km) are evaluated. The algorithm selects B because $h(B) < h(A)$, ignoring whether actual road distance $g(S \rightarrow B)$ is significantly longer than $g(S \rightarrow A)$.

b) Describe the architecture of knowledge based agent in AI.

A Knowledge-Based Agent uses an internal formal representation of knowledge to reason about world states and select optimal actions.



- **Core Components:**
 - **Knowledge Base (KB):** A centralized repository of formal sentences (facts and rules) expressed in a knowledge representation language.
 - **Inference Engine:** Reasoning mechanisms (Forward/Backward Chaining, Resolution) that derive new assertions from existing knowledge.
- **Agent Operational Cycle:**

- 1. TELL(KB, perceive(t)): Informs KB of current sensory inputs.
- 2. ASK(KB, action_query(t)): Queries inference engine to select best action.
- 3. TELL(KB, perform(action, t)): Updates KB to log executed action.

c) Elaborate data cleaning with example.

Data cleaning is the data preprocessing phase focused on identifying, correcting, or removing inaccurate, incomplete, duplicate, corrupted, or missing data points from raw datasets.

- **Key Cleaning Steps:** Missing value imputation, deduplication, outlier treatment, and format standardization.

Concrete Example:

Raw Input Data:

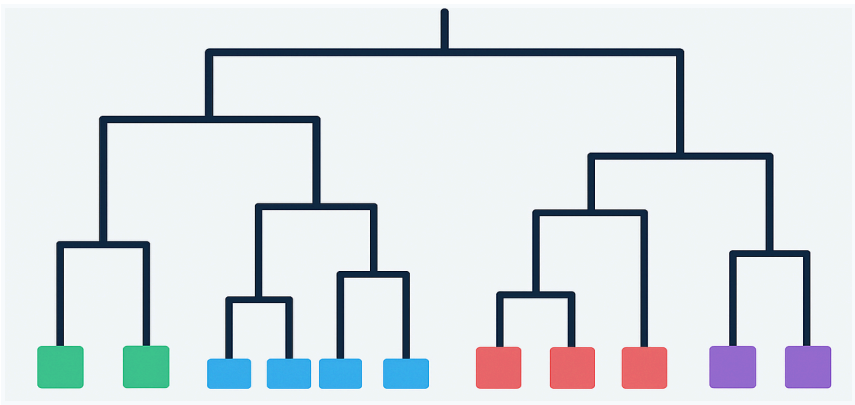
Customer_ID	Age	Location	Purchase_Amount
101	28	"NY"	\$150
102	NaN	"New York"	\$200
101	28	"NY"	\$150

Cleaning Operations:

- 1. Remove duplicate entry for Customer_ID 101.
- 2. Impute missing Age for Customer_ID 102 using median age (28).
- 3. Standardize Location string values ("New York" → "NY").

d) Explain Hierarchical clustering unsupervised learning algorithm.

Hierarchical clustering organizes data points into a continuous tree structure of nested clusters called a **Dendrogram**, eliminating the need to pre-specify cluster count *K*.



- **Main Approaches:**
 - **Agglomerative (Bottom-Up):** Starts with each point in its own cluster and

iteratively merges closest pairs until a single root cluster remains.

- **Divisive (Top-Down):** Starts with all points in one cluster and recursively splits them into smaller sub-clusters.
- **Linkage Methods:** Single Linkage (minimum distance), Complete Linkage (maximum distance), Average Linkage, Ward's Linkage (minimizes variance).

e) Differentiate between overfitting and underfitting.

Feature	Overfitting	Underfitting
Definition	Model learns noise and specific training details excessively, failing to generalize to unseen data.	Model is overly simplistic and fails to capture underlying data relationships.
Bias / Variance	Low Bias, High Variance .	High Bias , Low Variance.
Accuracy Profile	High training accuracy, low testing accuracy.	Low training accuracy, low testing accuracy.
Causes	Excessive model complexity, high parameter count, training for too many epochs.	Oversimplified model structure, insufficient features, over-regularization.
Remedies	Apply regularization (L1/L2), cross-validation, pruning, dropout, increase training data.	Increase model complexity, engineer relevant features, reduce regularization penalties.

5. Attempt any TWO of the following (12 Marks)

a) Explain Intelligent agent in AI w.r.t. structure of agent and Turing test in AI.

1. Structure of an Intelligent Agent:

An intelligent agent interacts with its environment by receiving percepts through sensors and executing actions via actuators.

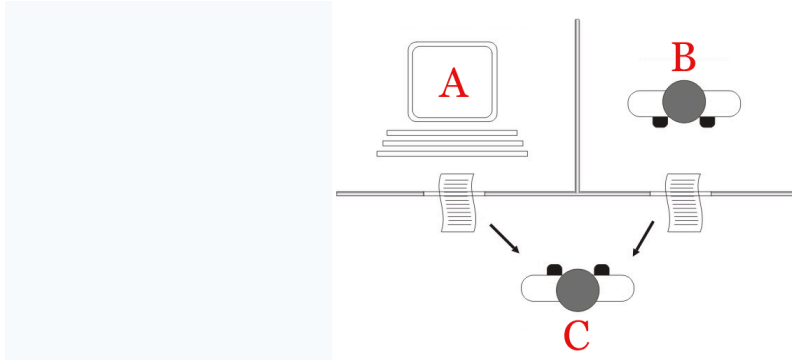
Agent Structure = Architecture + Agent Program

- **Architecture:** The physical computing system hardware executing the agent.
- **Agent Program:** The software function mapping percept sequences to actions: $f: P^* \rightarrow A$.
- **Key Components:**
 - **Sensors:** Perceptual inputs (cameras, sensors, input streams).

- **Actuators:** Mechanisms executing physical or digital actions (motors, display screens, API calls).

2. The Turing Test in AI:

Proposed by Alan Turing in 1950, the Turing Test establishes a benchmark for assessing human-equivalent machine intelligence.

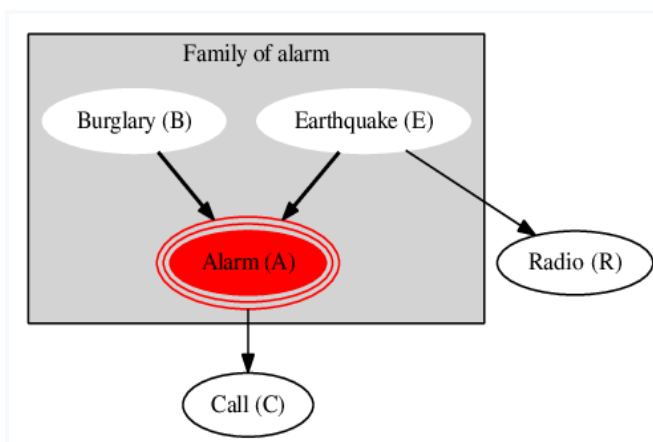


- **Setup:** An interrogator communicates blindly via text with a human and a computer. If the interrogator cannot reliably distinguish machine from human after evaluation, the machine passes.
- **Required Capabilities:** Natural Language Processing (NLP), Knowledge Representation, Automated Reasoning, and Machine Learning.

b) Illustrate probabilistic reasoning in AI with examples.

Probabilistic reasoning allows AI systems to make decisions under uncertainty, noise, and incomplete knowledge using probability theory.

Bayesian Network Example (Burglar Alarm System):



- **Prior Probabilities:** $P(B) = 0.001$, $P(E) = 0.002$.
- **Conditional Probabilities:** $P(A | B, \neg E) = 0.94$, $P(J | A) = 0.90$, $P(M | A) = 0.70$.
- **Reasoning Process:** When John calls ($J = \text{True}$), the system computes updated belief state $P(B | J)$ by marginalizing over intermediate Alarm node A . If Mary also calls ($M = \text{True}$), joint evidence $P(B | J, M)$ updates dynamically using Bayesian Inference.

c) Explain metrics for Regression: **MSE, RMSE, MAE.**

- **i) Mean Squared Error (MSE):** Computes the average of squared errors between actual target values (y_i) and predicted values ($\hat{y}_i$):

$$\text{MSE} = (1 / n) \sum (y_i - \hat{y}_i)^2$$
 Penalizes large errors heavily due to the squaring operation.
- **ii) Root Mean Squared Error (RMSE):** Square root of MSE, returning error scale back to original target units:

$$\text{RMSE} = \sqrt{[(1 / n) \sum (y_i - \hat{y}_i)^2]}$$
 Highly sensitive to severe outliers while remaining easy to interpret.
- **iii) Mean Absolute Error (MAE):** Computes average absolute differences between ground truth targets and predictions:

$$\text{MAE} = (1 / n) \sum |y_i - \hat{y}_i|$$
 Penalizes all errors linearly; robust against extreme outliers.

6. Attempt any TWO of the following (12 Marks)

a) Explain different forms of data.

1. Structural Forms of Data:

- **Structured Data:** Highly organized in tabular rows and columns with fixed schemas (e.g., SQL tables, CSV spreadsheets).
- **Semi-Structured Data:** Lacks rigid relational structure but contains organizational tags or key-value hierarchies (e.g., JSON, XML, HTML).
- **Unstructured Data:** Raw, unformatted data representing over 80% of real-world enterprise information (e.g., audio, video, images, free-text documents).

2. Statistical Data Types:

- **Nominal Data:** Categorical features without natural ordering (e.g., Gender, Color).
- **Ordinal Data:** Categorical features with inherent sequential rank (e.g., High / Medium / Low).
- **Interval Data:** Numeric scale with equal intervals but no true zero point (e.g., Temperature in °C).
- **Ratio Data:** Numeric scale with equal intervals and an absolute zero point (e.g., Weight, Income, Distance).

b) Differentiate between supervised and unsupervised learning.

Feature	Supervised Learning	Unsupervised Learning
Input Data	Requires labeled datasets (input features X paired with target ground truth Y).	Operates on unlabeled raw datasets (feature matrix X without target labels).

Feature	Supervised Learning	Unsupervised Learning
Primary Goal	Predict target labels or continuous values for new unseen input instances.	Discover inherent patterns, clusters, structures, or lower-dimensional representations.
Feedback Mechanism	Explicit error feedback derived from comparison against true target labels.	No explicit external feedback or supervisory guidance.
Task Types	Classification and Regression tasks.	Clustering, Dimensionality Reduction, and Association Rule Mining.
Algorithms	Linear Regression, Logistic Regression, Decision Trees, SVM.	K-Means, Hierarchical Clustering, PCA, Apriori.
Evaluation	Direct evaluation using ground truth metrics (Accuracy, F1-Score, MSE).	Intrinsic evaluation metrics (Silhouette Score, Inertia, Within-Cluster Variance).

c) Implement multiple linear regression with example.

Multiple Linear Regression models the linear relationship between a continuous scalar target variable Y and multiple feature variables $X_1, X_2, \dots, X_p$:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon$$

Python Code Implementation:

```
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score

# 1. Create Sample Dataset (House Price Prediction)
data = {
    'Area_SqFt': [1200, 1500, 1800, 2400, 3000, 3500],
    'Bedrooms': [2, 3, 3, 4, 4, 5],
    'Age_Years': [15, 10, 8, 5, 3, 1],
    'Price': [200000, 250000, 310000, 420000, 530000, 620000]
}
df = pd.DataFrame(data)
```

```
# 2. Separate Features (X) and Target (y)
X = df[['Area_SqFt', 'Bedrooms', 'Age_Years']]
y = df['Price']

# 3. Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2, random_state=42)

# 4. Instantiate and Train Linear Regression Model
model = LinearRegression()
model.fit(X_train, y_train)

# 5. Predict on Test Set and Evaluate Performance
y_pred = model.predict(X_test)

print("Intercept (Beta 0):", model.intercept_)
print("Coefficients (Beta 1, Beta 2, Beta 3):", model.coef_)
print("Mean Squared Error (MSE):", mean_squared_error(y_test,
y_pred))
print("R-squared Score (R2):", r2_score(y_test, y_pred))
```